```
!python --version
```

```
Python 3.12.13
```

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
```

```
import pandas as pd
```

Q1. Load the given retail sales dataset into a Pandas DataFrame and display the first 5 rows along with the dataset information to understand its structure.

```
csv_data= pd.read_csv("/content/retail_dataset  (1).csv")
```

```
type(csv_data)
```

```
pandas.core.frame.DataFrame
def __init__(data=None, index: Axes | None=None, columns: Axes | None=None, dtype: Dtype |
None=None, copy: bool | None=None) -> None
```

```
Two-dimensional, size-mutable, potentially heterogeneous tabular data.

Data structure also contains labeled axes (rows and columns).
Arithmetic operations align on both row and column labels. Can be
thought of as a dict-like container for Series objects. The primary
pandas data structure.
```

```
csv_data.head()
```

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region
0	T001	C015	Books	7.0	167.0	02-02-2023	North
1	T002	C004	Toys	4.0	374.0	17-02-2023	South
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South
4	T005	C022	Books	7.0	282.0	27-07-2024	West

Q2. Inspect the dataset and identify missing values in each column. Display the total number of missing values present in the dataset.

```
missing = csv_data.isnull().sum()
print(missing)
print("Total missing values:", missing.sum())
```

```
Transaction ID      0
Customer ID         0
Product Category    3
Quantity            2
Price               5
Date                0
Region              2
dtype: int64
Total missing values: 12
```

```
import pandas as pd
import numpy as np
```

Q3. Replace missing values in the Quantity and Price columns with their respective mean values using Pandas.

```
csv_data['Quantity'] = csv_data['Quantity'].fillna(csv_data['Quantity'].mean())
csv_data['Price'] = csv_data['Price'].fillna(csv_data['Price'].mean())
print(csv_data[['Quantity','Price']].mean())
```

```
Quantity      5.959184
Price       820.526316
dtype: float64
```

Q4. Remove all rows where the Product Category or Region column contains missing values.

```
df_cleaned = csv_data.dropna(subset=['Product Category', 'Region'])

print(f"Original Dataset Rows: {len(csv_data)}")
print(f"Cleaned Dataset Rows : {len(df_cleaned)}")
```

```
Original Dataset Rows: 100
Cleaned Dataset Rows : 95
```

Q5. Create a new column called Revenue using NumPy, where: Revenue = Quantity * Price Display the updated dataset with the new column.

```
import numpy as np

csv_data['Revenue'] = np.multiply(csv_data['Quantity'].values, csv_data['Price'].values)

print(csv_data.head())
```

```
  Transaction ID Customer ID Product Category  Quantity   Price        Date  \
0           T001        C015            Books       7.0   167.0  02-02-2023
1           T002        C004             Toys       4.0   374.0  17-02-2023
2           T003        C033        Furniture       8.0  1059.0  17-05-2024
3           T004        C047        Groceries       5.0   304.0  11-02-2024
4           T005        C022            Books       7.0   282.0  27-07-2024

  Region  Revenue
0  North   1169.0
1  South   1496.0
2  South   8472.0
3  South   1520.0
4   West   1974.0
```

Q6. Using NumPy, calculate the total revenue generated from all transactions in the dataset.

```
total_revenue = np.nansum(csv_data['Quantity'].values * csv_data['Price'].values)

print("Total Revenue :", total_revenue)
```

```
Total Revenue : 462851.0
```

Q7. Group the dataset by Product Category and calculate the total revenue generated by each category.

```
csv_data['Revenue'] = csv_data['Quantity'] * csv_data['Price']
category_revenue = (csv_data.groupby('Product Category')['Revenue']
    .sum()
    .reset_index()
    .sort_values(by='Revenue', ascending=False)
)

print(category_revenue)
```

```
  Product Category   Revenue
4        Furniture  128844.0
2         Clothing   85006.0
3      Electronics   83358.0
7             Toys   45270.0
6           Sports   34484.0
0           Beauty   28691.0
1            Books   26523.0
5        Groceries   11978.0
```

Q8. Based on the revenue calculated for each product category, identify the top 3 categories wit the highest revenue and the bottom 3 categories with the lowest revenue.

```
top_3_categories = category_revenue.head(3)
bottom_3_categories_list =
```

```
top_3_categories
```

	Product Category	Revenue
4	Furniture	128844.0
2	Clothing	85006.0
3	Electronics	83358.0

```
bottom_3_categories_list
```

	Product Category	Revenue
0	Beauty	28691.0
1	Books	26523.0
5	Groceries	11978.0

Q9. Group the dataset by Region and calculate the total revenue generated in each region. Identify the region with the highest revenue and the region with the lowest revenue.

```
csv_data['Revenue'] = csv_data['Quantity'] * csv_data['Price']

region_revenue = csv_data.groupby('Region')['Revenue'].sum().sort_values(ascending=False)
print(region_revenue)
print("Highest:", region_revenue.idxmax(), "| Lowest:", region_revenue.idxmin())
```

```
Region
North    155132.0
East     128827.0
South     97471.0
West      80377.0
Name: Revenue, dtype: float64
Highest: North | Lowest: West
```

Q10. Extract the month from the Date column using Pandas and group the dataset by month to calculate the total monthly revenue. Also, calculate the mean, median, and standard deviation of the Revenue column using NumPy.

```
csv_data['Date'] = pd.to_datetime(csv_data['Date'], format='%d-%m-%Y')
csv_data['Month'] = csv_data['Date'].dt.month
monthly_revenue = csv_data.groupby('Month')['Revenue'].sum()
print(monthly_revenue)

mean_rev = np.nanmean(csv_data['Revenue'].values)
median_rev = np.nanmedian(csv_data['Revenue'].values)
std_rev = np.nanstd(csv_data['Revenue'].values)

print(f"Mean: {mean_rev:.2f} | Median: {median_rev:.2f} | Std Dev: {std_rev:.2f}")
```

```
Month
1     43994.0
2     33028.0
3     10557.0
4     56478.0
5     27915.0
6     37935.0
7     36919.0
8     76958.0
9     27437.0
10    42097.0
11    25150.0
12    44383.0
Name: Revenue, dtype: float64
Mean: 4976.89 | Median: 3552.00 | Std Dev: 4764.81
```